CaSDOM: Contactless Sleep Disorder Monitoring Using Radar Sensing

Furui Chen, Zhi Lu, Fang Zhou, Yu Pu, Dongheng Zhang, and Yan Chen
School of Cyberspace Security, University of Science and Technology of China (USTC), Hefei, China
chenfurui@mail.ustc.edu.cn, zhilu@ustc.edu.cn, fangzhou1958@mail.ustc.edu.cn,
puyu@mail.ustc.edu.cn, dongheng@ustc.edu.cn, eecyan@ustc.edu.cn

***Abstract*—Accurate and low-burden overnight assessment of obstructive sleep apnea/hypopnea syndrome (OSAHS) is essential for large-scale screening and long-term monitoring. Radar-only OSAHS assessment is challenging because radar does not directly measure airflow, and radar-derived respiratory proxies can be sensitive to multipath and body motion. To overcome this challenge, we propose CaSDOM, a radar-only framework for overnight respiratory event detection and AHI-based severity assessment. CaSDOM constructs two complementary respiratory proxies via array-based spatial filtering and coherent multi-cell combining, and incorporates hour-scale temporal context with a bidirectional structured state-space model to reduce the impact of short-term proxy instability on event recognition. Evaluated on 100 PSG–Radar synchronized overnight recordings using subject-wise, AHI-stratified five-fold cross-validation, CaSDOM achieves a Macro-F1 of 0.732 and an AHI MAE of 4.02 events/hour with r = 0.974, outperforming representative baselines.**

***Index Terms*—array processing, beamforming, contactless sleep monitoring, millimeter-wave radar, long-context temporal modeling**

I. Introduction

Obstructive sleep apnea/hypopnea syndrome (OSAHS) is a prevalent sleep-related breathing disorder characterized by recurrent upper-airway obstruction during sleep, leading to repeated apnea and hypopnea events [1], [2]. Disease severity is quantified by the apnea-hypopnea index (AHI) and stratified into normal, mild, moderate, and severe categories [3]. OSAHS fragments sleep and increases cardiovascular and metabolic risks via intermittent hypoxia and sympathetic activation [4]–[6]. Accurate, low-burden assessment is essential for large-scale screening and long-term monitoring.

Polysomnography (PSG) remains the clinical gold standard, but body-attached sensors and expert supervision limit scalable home use [3], [7], [8]. This motivates contactless sleep monitoring. Millimeter-wave frequency-modulated continuous-wave radar is privacy-preserving and lighting-invariant, sensing thoracoabdominal micro-motions related to respiration [9]–[11]. However, reliable radar-only overnight OSAHS assessment in real bedrooms remains challenging.

A fundamental challenge in radar-only OSAHS assessment is that radar does not directly measure airflow. Respiration must be inferred from far-field reflections, making radar-derived proxies sensitive to multipath, clutter, body motion, and posture/orientation changes [12]–[14]. As a result, proxy quality varies substantially across subjects and nights, and representations based on short temporal windows are often unstable, particularly during motion transitions.

To address this challenge, we propose CaSDOM, a radar-only framework for overnight OSAHS event detection and AHI-based severity assessment. CaSDOM performs digital beamforming on the TDM-MIMO virtual array and constructs two complementary respiratory proxies using energy- and spectral-based criteria. It then aggregates patch-level features with hour-scale temporal context using a bidirectional structured state-space model (Bi-S4D), reducing the impact of short-term observation instability on event recognition. On 100 overnight PSG–Radar recordings with subject-wise, AHI-stratified five-fold cross-validation, CaSDOM achieves a Macro-F1 of 0.732 and an AHI MAE of 4.02 events/hour, with $r = 0.974$.

The main contributions are summarized as follows:

- We propose CaSDOM, a radar-only framework that constructs two complementary respiratory proxies *via array-based spatial filtering* for robust sensing in real bedrooms.
- We introduce efficient hour-scale event modeling that integrates local event morphology with long-range temporal context using a bidirectional structured state-space backbone (Bi-S4D).
- We evaluate CaSDOM on 100 overnight PSG–Radar recordings under subject-wise, AHI-stratified five-fold cross-validation, achieving a Macro-F1 of 0.732 and an AHI MAE of 4.02 events/hour with $r = 0.974$, outperforming representative baselines.

II. Methods

Fig. 1 provides an overview of CaSDOM. Our method is organized into two stages: (i) preprocessing for label alignment and dual-proxy construction, and (ii) hour-scale long-context modeling for respiratory event detection and AHI-based severity assessment. The two stages are described in the following subsections.

A. Signal Preprocessing

1) Label Alignment for Training and Evaluation: Radar recordings are aligned with PSG annotations by synchronizing acquisition server timestamps. This step is used only to obtain labels for training and evaluation, while CaSDOM uses only radar signals during inference.

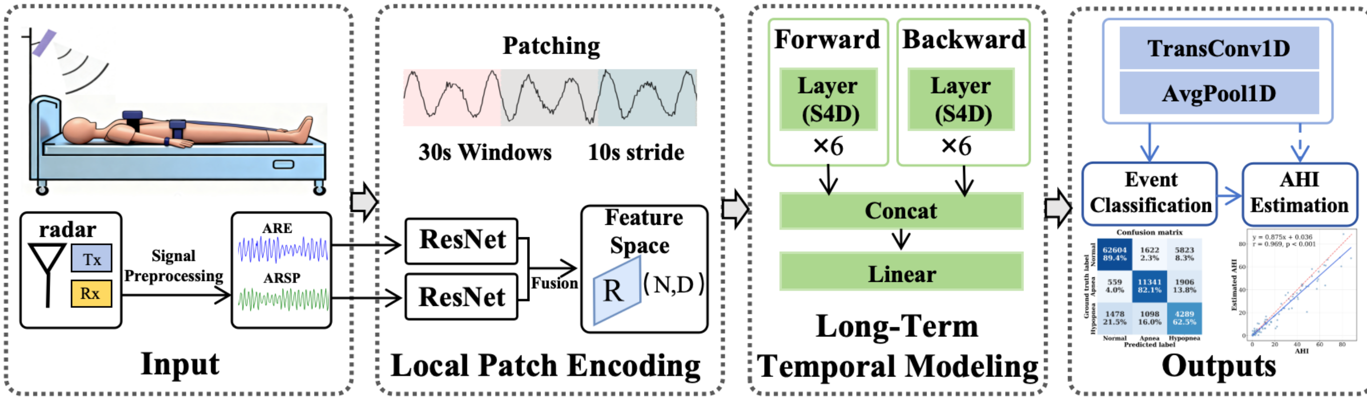

Fig. 1: Overall framework of CaSDOM, consisting of: (1) preprocessing for label alignment and dual-proxy construction; and (2) hour-scale long-context modeling for respiratory event detection and AHI-based severity assessment.

2) Spatial Filtering and Dual-Proxy Construction: We use a Texas Instruments AWR6843 FMCW millimeter-wave radar operating at 60 GHz with a 3-Tx/4-Rx TDM-MIMO array, forming 12 virtual channels. The radar platform provides an approximately 120° field of view, a maximum transmission power of 15 dBm, a maximum sweep bandwidth of 4 GHz, and a frame rate of 100 Hz. To enhance spatial selectivity, we perform digital beamforming and obtain complex returns on a discrete range–azimuth grid [15]–[17].

Each range–azimuth cell p yields a candidate phase trace $\phi_p[m]$ from the unwrapped phase. Let $P_p(f)$ denote its power spectrum. We score each cell using a motion-energy score and a breathing-band ratio:

$$E_p = \frac{1}{M}\sum_{m=1}^{M} |\phi_p[m]|^2, \quad \mathrm{BNR}_p = \frac{\max_{f\in\mathcal{B}} P_p(f)}{\sum_{f\in\Omega} P_p(f)}, \tag{1}$$

where $\mathcal{B} = [0.15, 0.5]$ Hz denotes the respiratory band and Ω denotes the valid frequency bins. The top-K cells ranked by E_p and BNR_p are coherently combined via reference-cell phase compensation and summation, producing the energy-guided proxy **ARE** and the spectrum-guided proxy **ARSP**, respectively.

3) Temporal Filtering and Downsampling: The resulting ARE and ARSP proxies are lightly smoothed to suppress high-frequency fluctuations and then downsampled from 100 Hz to 10 Hz. The processed proxies are used as synchronized two-channel inputs to the downstream model.

B. Model

1) Local Patch Encoding: Sleep physiological signals have high sampling rates and long monitoring durations. Directly modeling an overnight recording is computationally expensive and may dilute short-term cues for respiratory event detection. We therefore adopt a patch-based representation: the two array-derived proxy sequences (ARE/ARSP) are divided into overlapping short-time patches and encoded into compact tokens for hour-scale temporal modeling.

Let the sampling rate be F_s=10 Hz and the window length be T=3600 s. Denote the two proxy signals as $\mathbf{a}(t)$ (ARE) and $\mathbf{p}(t)$ (ARSP). We apply a sliding window with patch length P=30 s and stride S=10 s, yielding the patch count

$$N = \left\lfloor \frac{T-P}{S} \right\rfloor + 1. \tag{2}$$

In practice, we pad to handle the last patch. The n-th patch is denoted by $\mathbf{a}^{(n)}, \mathbf{p}^{(n)} \in \mathbb{R}^{PF_s}$ for $n = 1, \dots, N$.

To capture channel-specific morphology, we employ two unshared 1D ResNet-34 encoders [20]:

$$\mathbf{h}_n^a = f_a(\mathbf{a}^{(n)}), \quad \mathbf{h}_n^p = f_p(\mathbf{p}^{(n)}) \in \mathbb{R}^D, \quad D = 512, \tag{3}$$

where $f_a(\cdot)$ and $f_p(\cdot)$ share the same architecture but do not share weights. We fuse the dual-proxy embeddings via an MLP $\Phi : \mathbb{R}^{2D} \to \mathbb{R}^D$:

$$\mathbf{r}_n = \Phi([\mathbf{h}_n^a; \mathbf{h}_n^p]), \qquad \mathbf{R} = [\mathbf{r}_1, \dots, \mathbf{r}_N]^\top \in \mathbb{R}^{N\times D}. \tag{4}$$

2) Long-Term Temporal Modeling: While $\mathbf{R} \in \mathbb{R}^{N\times D}$ captures local patterns, robust respiratory event detection also requires modeling hour-scale dependencies, such as clustered events and gradual physiological transitions. To capture long-range context efficiently, we adopt a bidirectional structured state-space model (Bi-S4D) based on the diagonal S4D variant [21], which enables linear-time sequence modeling in N (avoiding the $O(N^2)$ cost of self-attention).

Respiratory event detection benefits from both historical and future context. We apply the same S4D encoder on the forward and reversed token sequences to obtain $\mathbf{h}_n^{\rightarrow}$ and $\mathbf{h}_n^{\leftarrow}$, and fuse them via a linear projection:

$$\mathbf{H}_n = \mathbf{U}\,[\mathbf{h}_n^{\rightarrow}; \mathbf{h}_n^{\leftarrow}], \qquad \mathbf{H} = [\mathbf{H}_1, \dots, \mathbf{H}_N]^\top \in \mathbb{R}^{N\times D}, \tag{5}$$

where $\mathbf{U} \in \mathbb{R}^{D\times 2D}$ is learnable.

3) Multi-Scale Outputs: Given the hour-scale latent sequence $\mathbf{H} \in \mathbb{R}^{N\times D}$, CaSDOM outputs class-score sequences at two temporal scales: a second-level trajectory at 1 Hz resolution and an epoch-level sequence aligned with clinical scoring. Let T = 3600 s, L_{ep} = 30 s, $E = T/L_{\mathrm{ep}}$, and C = 3 for normal, apnea, and hypopnea. **Second-level outputs.** Second-level class scores are obtained by upsampling $\mathbf{H}$ to 1 Hz

TABLE I: Overall performance on PRS (five-fold mean ± std). Best results are in bold. (↑: higher is better, ↓: lower is better)

Method	Epoch-level Event Classification			Overnight AHI Evaluation			
	Acc↑	Macro-F1↑	κ ↑	MAE↓	r ↑	Acc_{sev} ↑	κ_{sev} ↑
Vaquerizo-Villar et al. [18]	0.835±0.017	0.599±0.033	0.478±0.043	5.44±0.67	0.968±0.015	0.750±0.077	0.654±0.108
Choi et al. [10]	0.851±0.014	0.625±0.013	0.537±0.029	5.57±0.38	0.971±0.011	0.740±0.073	0.635±0.108
Zhuang et al. [19]†	–	–	–	7.29	0.932	0.650	0.535
Ours	**0.862±0.008**	**0.732±0.011**	**0.668±0.017**	**4.02±0.60**	**0.974±0.018**	**0.850±0.055**	**0.796±0.074**

† Results are obtained from a single test-only evaluation on PRS without cross-validation; epoch-level metrics and standard deviations are not reported.

resolution using a transposed convolution $\text{TConv}_{P,S}(\cdot)$ with kernel size P and stride S (both in seconds):

$$\hat{\mathbf{Y}}_{\text{sec}} = \text{TConv}_{P,S}(\mathbf{H}) \in \mathbb{R}^{T\times C}. \tag{6}$$

Epoch-level outputs. Epoch-level scores are computed by average pooling over non-overlapping L_{ep}-second intervals:

$$\hat{\mathbf{Y}}_{\text{ep}} = \text{AvgPool}_{L_{\text{ep}}}(\hat{\mathbf{Y}}_{\text{sec}}) \in \mathbb{R}^{E\times C}. \tag{7}$$

4) Learning Objectives: Let $y_{\text{ep}}(e) \in \{0,1,2\}$ denote the ground-truth label of epoch e (0: normal, 1: apnea, 2: hypopnea), and let $y_{\text{sec}}(t) \in \{0,1,2\}$ denote the second-level ground-truth label at second t. We use $\hat{\mathbf{Y}}_{\text{ep}} \in \mathbb{R}^{E\times C}$ and $\hat{\mathbf{Y}}_{\text{sec}} \in \mathbb{R}^{T\times C}$ to denote the corresponding predicted class-score sequences. Training is driven by epoch-level three-class detection, with an additional normal–abnormal constraint and an hourly event-count consistency term based on second-level aggregation.

Epoch-level three-class detection. We supervise $\hat{\mathbf{Y}}_{\text{ep}}$ using class-weighted cross-entropy:

$$\mathcal{L}_{\text{ep}} = \text{CE}_w(\hat{\mathbf{Y}}_{\text{ep}}, y_{\text{ep}}). \tag{8}$$

where $\text{CE}_w(\cdot)$ denotes cross-entropy with class weights computed from training-set label frequencies.

Normal–abnormal constraint. Following the American Academy of Sleep Medicine (AASM) scoring rules, apnea and hypopnea are both abnormal events, while their subtype boundary depends on threshold-based criteria (e.g., desaturation/arousal) and can be ambiguous near decision margins. To stabilize learning, we add an auxiliary normal–abnormal loss by marginalizing the three-class prediction into a binary abnormal probability $p_{\text{abn}} = \hat{p}_{apnea} + \hat{p}_{hypopnea}$ and applying cross-entropy with target $y_{\text{bin}}(e) = \mathbf{1}[y_{\text{ep}}(e) > 0]$. We denote this auxiliary loss as $\mathcal{L}_{\text{bin}} = \text{CE}([1 - p_{\text{abn}},\, p_{\text{abn}}], y_{\text{bin}})$.

Hourly event-count consistency. For severity assessment, we regress the hourly event count from the second-level trajectory. Under AASM scoring, an event is a contiguous abnormal segment lasting at least 10 s; the hourly count is denoted by N_{AH}. We predict $\hat{N}_{AH} = g(\hat{\mathbf{Y}}_{\text{sec}})$ via AASM-compliant aggregation (≥10 s) and minimize the mean-squared error:

$$\mathcal{L}_{\text{cnt}} = \|\hat{N}_{AH} - N_{AH}\|_2^2. \tag{9}$$

Total objective. The final objective is

$$\mathcal{L} = \lambda_{\text{ep}}\mathcal{L}_{\text{ep}} + \lambda_{\text{bin}}\mathcal{L}_{\text{bin}} + \lambda_{\text{cnt}}\mathcal{L}_{\text{cnt}}, \tag{10}$$

where we set $\lambda_{\text{ep}} = 1.0$, $\lambda_{\text{bin}} = 1.0$, and $\lambda_{\text{cnt}} = 2 \times 10^{-4}$ in this study.

AHI derivation. For an overnight recording, we sum predicted hourly event counts and compute $\widehat{\text{AHI}} = \sum_h \hat{N}_{AH}^{(h)}/\text{TST}_{\text{hours}}$, where $\text{TST}_{\text{hours}}$ denotes total sleep time (from PSG) in hours.

III. Experiments

A. Experimental Setup

1) Datasets: Data annotation follows AASM standards. We construct a PSG–Radar Synchronized (PRS) dataset by simultaneously recording PSG and millimeter-wave radar signals at a tertiary hospital. The dataset includes 100 overnight recordings from 77 subjects, totaling approximately 900 hours, with 33 normal (AHI< 5), 19 mild (5≤AHI< 15), 15 moderate (15≤AHI< 30), and 33 severe (AHI≥30) recordings.

2) Training and Evaluation Protocol: We pretrain the model on the Sleep Heart Health Study (SHHS) dataset [22], [23] using contact-based abdominal and thoracic respiratory belt signals (ABD/THO) with the Adam optimizer and a cosine learning-rate schedule (lr $= 1 \times 10^{-4}$), and fine-tune it on PRS with lr $= 5 \times 10^{-5}$. We conduct subject-wise, AHI-stratified five-fold cross-validation on PRS (20 nights per fold; no subject overlap), train up to 250 epochs with early stopping (patience = 20), and report mean±std over folds.

3) Compared Methods: We compare CaSDOM with three baselines: (i) Vaquerizo-Villar et al. [18], a SpO_2-based 1D-CNN *contact-based baseline* for three-class detection and AHI estimation; (ii) Choi et al. [10], a dual-radar CNN–Transformer for detection and severity grading; (iii) Zhuang et al. [19], a belt-pretrained residual CNN transferred to radar for AHI estimation (test-only on PRS with released weights).

4) Evaluation Metrics: For epoch-level three-class classification, we report accuracy (Acc), macro-averaged F1 (Macro-F1), and Cohen's kappa (κ). For overnight AHI/severity evaluation, we report mean absolute error of AHI (MAE), Pearson correlation coefficient (r), AHI severity classification accuracy (Acc_{sev}), and Cohen's kappa for severity (κ_{sev}).

B. Experimental Results

1) Overall Model Performance: Table I summarizes performance on PRS, where CaSDOM achieves the best results for epoch-level event classification and overnight AHI evaluation.

Epoch-level respiratory event detection. For 30-second three-class classification, CaSDOM attains the highest accuracy (0.862), Macro-F1 (0.732), and Cohen's κ (0.668),

TABLE II: Ablation study on PRS (five-fold mean ± std). Best results are in bold. (↑: higher is better, ↓: lower is better)

Variant	Epoch-level Event Classification			Overnight AHI Evaluation			
	Acc↑	Macro-F1↑	κ ↑	MAE↓	r ↑	Acc_{sev} ↑	κ_{sev} ↑
ARE-only	0.857±0.012	0.687±0.019	0.635±0.033	4.93±0.48	0.967±0.019	0.750±0.110	0.655±0.152
ARSP-only	0.845±0.011	0.676±0.012	0.612±0.021	5.14±0.58	0.964±0.019	0.780±0.081	0.698±0.112
w/o N–A constraint	0.852±0.004	0.713±0.017	0.646±0.016	4.30±0.67	0.973±0.017	0.810±0.066	0.739±0.092
Bi-S4D (w/o pretraining)	0.847±0.009	0.669±0.025	0.614±0.023	5.08±0.46	0.961±0.024	0.800±0.077	0.725±0.106
LSTM (w/o pretraining)	0.847±0.012	0.668±0.011	0.607±0.022	5.25±0.34	0.965±0.026	0.760±0.037	0.671±0.053
Transformer (w/o pretraining)	0.851±0.007	0.660±0.009	0.607±0.019	5.49±0.18	0.961±0.023	0.750±0.100	0.657±0.136
Full CaSDOM	**0.862±0.008**	**0.732±0.011**	**0.668±0.017**	**4.02±0.60**	**0.974±0.018**	**0.850±0.055**	**0.796±0.074**

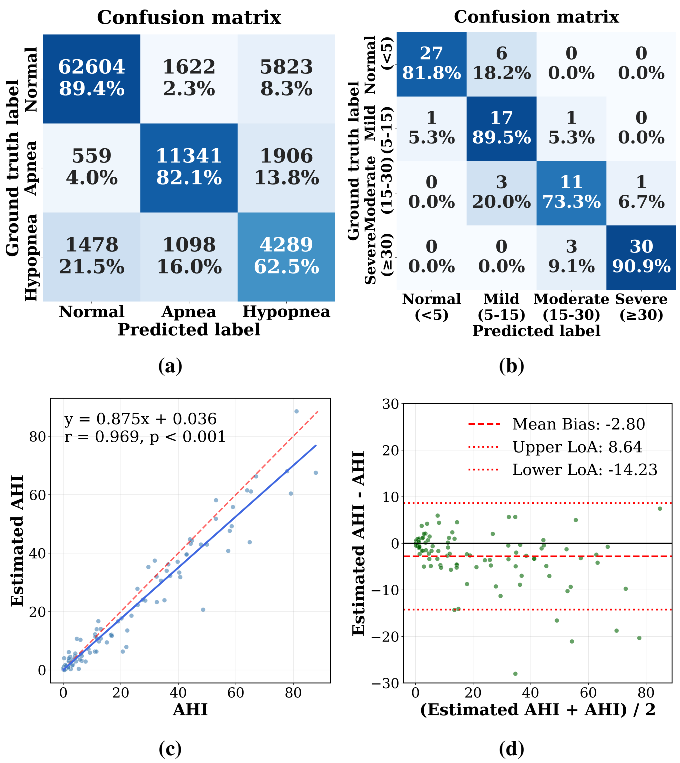

Fig. 2: Comprehensive evaluation of CaSDOM on the PRS dataset. Results are aggregated over all test folds. (a) Epoch-level event confusion matrix. (b) AHI severity classification confusion matrix. (c) Correlation between estimated and reference AHI values. (d) Bland–Altman analysis of agreement between estimated and reference AHI.

indicating improved consistency and balanced performance across event categories. As shown in Fig. 2(a), most residual errors arise between normal breathing and hypopnea, while apnea detection remains comparatively stable.

Overnight AHI estimation and severity classification. CaSDOM achieves the lowest AHI estimation error (MAE = 4.02), the highest correlation with reference AHI ($r = 0.974$), and the best severity-level accuracy (0.850) and κ_{sev} (0.796). As shown in Fig. 2(b–d), severity errors are mainly concentrated between adjacent AHI categories, and the estimated AHI shows strong agreement with the reference AHI.

2) Ablation Study: Table II reports controlled ablations on (i) dual-proxy inputs, (ii) the normal–abnormal (N–A) constraint, and (iii) long-context temporal modeling, including backbone choices and transfer pretraining.

Dual-proxy inputs. Removing either ARE or ARSP underperforms the full model, with Macro-F1 decreasing from 0.732 to 0.687/0.676 and MAE increasing from 4.02 to 4.93/5.14. This indicates that ARE and ARSP provide complementary cues, and their fusion improves epoch-level event detection and AHI estimation.

Normal–abnormal constraint. Removing the N–A constraint leads to lower agreement, with epoch-level κ decreasing from 0.668 to 0.646 and severity-level κ_{sev} decreasing from 0.796 to 0.739, accompanied by an increase in MAE from 4.02 to 4.30. These results suggest that the coarse normal–abnormal signal serves as a stabilizing regularizer for three-class learning, which in turn benefits event aggregation and severity estimation.

Long-context modeling and transfer initialization. Training Bi-S4D without pretraining leads to performance degradation, with Macro-F1 decreasing from 0.732 to 0.669 and MAE increasing from 4.02 to 5.08. This highlights the importance of belt-based pretraining for contactless transfer on PRS. Under the same no-pretraining setting, Bi-S4D remains competitive with LSTM [24] and Transformer [25] backbones, supporting its suitability for modeling hour-scale respiratory dynamics.

IV. Conclusion

This paper presents CaSDOM, a radar-only framework for overnight respiratory event detection and AHI-based severity assessment. By constructing complementary radar respiratory proxies and incorporating hour-scale long-context modeling, CaSDOM reduces the impact of proxy instability caused by multipath and body motion. Results on 100 PSG–Radar synchronized overnight recordings show that CaSDOM achieves accurate epoch-level event classification and overnight AHI estimation, outperforming representative baselines. Ablation studies further confirm the contributions of dual-proxy inputs, the normal–abnormal constraint, and hour-scale long-context modeling. Future work will extend validation to larger cohorts and explore real-time deployment.

Compliance with Ethical Standards

All experimental procedures were reviewed and approved by the institutional review board, and written informed consent was obtained from all participants prior to data acquisition.

REFERENCES

[1] A. S. Jordan, D. G. McSharry, and A. Malhotra, “Adult obstructive sleep apnoea,” *The Lancet*, vol. 383, no. 9918, pp. 736–747, 2014.

[2] S. C. Veasey and I. M. Rosen, “Obstructive sleep apnea in adults,” *New England Journal of Medicine*, vol. 380, no. 15, pp. 1442–1449, 2019.

[3] R. B. Berry, R. Brooks, C. E. Gamaldo, S. M. Harding, R. M. Lloyd, C. L. Marcus, and B. V. Vaughn, “The aasm manual for the scoring of sleep and associated events,” *Rules, Terminology and Technical Specifications, Darien, Illinois, American Academy of Sleep Medicine*, vol. 176, p. 2012, 2012.

[4] J. M. Marin, S. J. Carrizo, E. Vicente, and A. G. Agusti, “Long-term cardiovascular outcomes in men with obstructive sleep apnoea-hypopnoea with or without treatment with continuous positive airway pressure: an observational study,” *The Lancet*, vol. 365, no. 9464, pp. 1046–1053, 2005.

[5] L. Xia, Z. Li, Z. Xie, Q. Zhang, M. Li, C. Zhang, and Y. Chen, “Obstructive sleep apnea and type 2 diabetes: a bidirectional mendelian randomization study,” *Zhonghua er bi yan hou tou Jing wai ke za zhi= Chinese Journal of Otorhinolaryngology Head and Neck Surgery*, vol. 58, no. 10, pp. 974–979, 2023.

[6] A. S. Gami, D. O. Hodge, R. M. Herges, E. J. Olson, J. Nykodym, T. Kara, and V. K. Somers, “Obstructive sleep apnea, obesity, and the risk of incident atrial fibrillation,” *Journal of the American College of Cardiology*, vol. 49, no. 5, pp. 565–571, 2007.

[7] H. Zheng, M. Sowers, D. J. Buysse, F. Consens, H. M. Kravitz, K. A. Matthews, J. F. Owens, E. B. Gold, and M. Hall, “Sources of variability in epidemiological studies of sleep using repeated nights of in-home polysomnography: Swan sleep study,” *Journal of clinical sleep medicine*, vol. 8, no. 1, pp. 87–96, 2012.

[8] V. K. Kapur, D. H. Auckley, S. Chowdhuri, D. C. Kuhlmann, R. Mehra, K. Ramar, and C. G. Harrod, “Clinical practice guideline for diagnostic testing for adult obstructive sleep apnea: an american academy of sleep medicine clinical practice guideline,” *Journal of clinical sleep medicine*, vol. 13, no. 3, pp. 479–504, 2017.

[9] Y. Zhang, Z. Zhuang, D. Xu, H. Zhao, and H. Hong, “Non-contact sleep apnea detection system based on data-driven methods using fmcw radar,” in *2023 IEEE MTT-S International Wireless Symposium (IWS)*. IEEE, 2023, pp. 1–3.

[10] J. W. Choi, D. L. Koo, D. H. Kim, H. Nam, J. H. Lee, S.-N. Hong, and B. Kim, “A novel deep learning model for obstructive sleep apnea diagnosis: hybrid cnn-transformer approach for radar-based detection of apnea-hypopnea events,” *Sleep*, vol. 47, no. 12, p. zsae184, 2024.

[11] T. Koda, S. Okumura, H. Taki, S. Hamada, H. Sunadome, S. Sato, K. Chin, and T. Sakamoto, “Noncontact detection of sleep apnea using radar and expectation-maximization algorithm,” *IEEE Sensors Journal*, 2024.

[12] M. Alizadeh, G. Shaker, J. C. M. De Almeida, P. P. Morita, and S. Safavi-Naeini, “Remote monitoring of human vital signs using mm-wave fmcw radar,” *IEEE Access*, vol. 7, pp. 54 958–54 968, 2019.

[13] V. P. Tran, A. A. Al-Jumaily, and S. M. S. Islam, “Doppler radar-based non-contact health monitoring for obstructive sleep apnea diagnosis: A comprehensive review,” *big data and cognitive computing*, vol. 3, no. 1, p. 3, 2019.

[14] M. Mercuri, G. Sacco, R. Hornung, P. Zhang, H. J. Visser, M. Hijdra, Y.-H. Liu, S. Pisa, B. van Liempd, and T. Torfs, “2-d localization, angular separation and vital signs monitoring using a siso fmcw radar for smart long-term health monitoring environments,” *IEEE Internet of Things Journal*, vol. 8, no. 14, pp. 11 065–11 077, 2021.

[15] M. A. Richards, *Fundamentals of radar signal processing*. Mcgraw-hill New York, 2005, vol. 1.

[16] Z. Guo, B. Zhang, J. Chen, Y. Hu, and Y. Chen, “Contactless fine-grained cardiac events detection and segmentation with radio frequency signals,” *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 9, no. 3, pp. 1–28, 2025.

[17] S. Kutty and D. Sen, “Beamforming for millimeter wave communications: An inclusive survey,” *IEEE communications surveys & tutorials*, vol. 18, no. 2, pp. 949–973, 2015.

[18] F. Vaquerizo-Villar, D. Álvarez, G. C. Gutiérrez-Tobal, C. Arroyo-Domingo, F. Del Campo, and R. Hornero, “Deep-learning model based on convolutional neural networks to classify apnea–hypopnea events from the oximetry signal,” in *Advances in the Diagnosis and Treatment of Sleep Apnea: Filling the Gap Between Physicians and Engineers*. Springer, 2022, pp. 255–264.

[19] Z. Zhuang, B. Xue, Q. An, H. Chu, Y. Zhang, R. Chen, J. Xu, N. Ding, X. Cui, E. Wang *et al.*, “Advancing sleep health equity through deep learning on large-scale nocturnal respiratory signals,” *Nature Communications*, vol. 16, no. 1, p. 9334, 2025.

[20] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.

[21] A. Gu, K. Goel, A. Gupta, and C. Ré, “On the parameterization and initialization of diagonal state space models,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 35 971–35 983, 2022.

[22] G.-Q. Zhang, L. Cui, R. Mueller, S. Tao, M. Kim, M. Rueschman, S. Mariani, D. Mobley, and S. Redline, “The national sleep research resource: towards a sleep data commons,” *Journal of the American Medical Informatics Association*, vol. 25, no. 10, pp. 1351–1358, 2018.

[23] S. F. Quan, B. V. Howard, C. Iber, J. P. Kiley, F. J. Nieto, G. T. O’Connor, D. M. Rapoport, S. Redline, J. Robbins, J. M. Samet *et al.*, “The sleep heart health study: design, rationale, and methods,” *Sleep*, vol. 20, no. 12, pp. 1077–1085, 1997.

[24] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[25] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, vol. 30. Red Hook, NY, USA: Curran Associates, Inc., 2017, pp. 5998–6008.